

LABORATORY WORK #2

Normalization and DDL

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1. Normalization

The original Booking_receipt table is not in 1NF because seat_numbers contains a list such as "12A, 12B". Because 1NF is violated, the table also cannot be in 2NF or 3NF.

Form	Problem
1NF	seat_numbers is not atomic.
2NF	After splitting seats, some columns depend only on booking_id or passport_number.
3NF	Flight and airport information has transitive dependencies.

1NF

One row must contain one passenger and one seat.

BOOKING_RECEIPT_1NF (booking_id [PK], passport_number [PK], passenger_full_name, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_number, ticket_price)

2NF

Passenger data and booking data are separated from the booking-passenger row.

PASSENGER_2NF (passport_number [PK], passenger_full_name)

BOOKING_2NF (booking_id [PK], flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name)

BOOKING_PASSENGER_2NF (booking_id [PK, FK], passport_number [PK, FK], seat_number, ticket_price)

3NF

Airline, airport and flight information is stored separately.

PASSENGER (passenger_id [PK], passport_number [UNIQUE], passenger_full_name)

AIRLINE (airline_id [PK], airline_name [UNIQUE])

AIRPORT (airport_id [PK], airport_name [UNIQUE], city)

FLIGHT (flight_id [PK], flight_number [UNIQUE], departing_airport_id [FK], arriving_airport_id [FK], airline_id [FK])

BOOKING (booking_id [PK], flight_id [FK])

TICKET (booking_id [PK, FK], passenger_id [PK, FK], seat_number, ticket_price)

The underlined columns are primary keys. Columns marked FK are foreign keys.

2. Functional dependencies

Dependency	Why it is used
(booking_id, passport_number) -> seat_number, ticket_price	Defines one ticket row.
passport_number -> passenger_full_name	Passenger data goes to PASSENGER.
booking_id -> flight_number	Flight belongs to the booking.
flight_number -> airline_name, departure_airport_name, arrival_airport_name	Flight details go to FLIGHT.
departure_airport_name -> departure_city	City goes to AIRPORT.
arrival_airport_name -> arrival_city	City goes to AIRPORT.
airline_id -> airline_name	Airline data goes to AIRLINE.

3. Flights table and 3NF

Primary key: flight_id.

flight_id -> sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, updated_at.

The table is in 3NF. Airport names and cities are stored in Airport, while airline data is stored in Airline. Therefore, no non-key attribute in Flights depends on another non-key attribute.

4. DDL implementation

The SQL creates all required tables, primary keys, NOT NULL constraints and foreign keys.

Required change	SQL result
Rename airline_info	airline
Rename booking.price	ticket_price
Change departing_gate	TEXT
Remove airline.info	Column dropped

Relationships. Passengers -> Security_check, Booking and Baggage_check; Booking -> Baggage_check, Baggage, Boarding_pass and Booking_flight; Flights -> Booking_flight; Airport -> Flights; Airline -> Flights.

5. Sample data

Five normalized booking rows were inserted.

Booking	Passenger	Flight	Route	Seat	Price
1001	Amina Sarsenova	101	Almaty-Astana	12A	45000.00
1002	Daniyar Tulegen	101	Almaty-Astana	12B	45000.00
1003	Sofia Lee	102	Astana-Shymkent	08C	32000.00
1004	Noah Kim	103	Almaty-Istanbul	21F	185000.00
1005	Aisha Omar	102	Astana-Shymkent	09A	33500.00

SQL screenshot 1

Query Tool - airport_db/postgres@localhost:5432 - pgAdmin 4

File Edit View Query Tools Help

Open Save Execute Stop Database: airport_db

SQLQuery1.sql - Core tables

```
1  -- PostgreSQL 16
2  CREATE TABLE airline_info (
3      airline_id INT PRIMARY KEY,
4      airline_code VARCHAR(30) NOT NULL UNIQUE,
5      airline_name VARCHAR(50) NOT NULL,
6      airline_country VARCHAR(50) NOT NULL,
7      created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
8      updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
9      info VARCHAR(50) NOT NULL
10 );
11
12 CREATE TABLE airport (
13     airport_id INT PRIMARY KEY,
14     airport_name VARCHAR(50) NOT NULL,
15     country VARCHAR(50) NOT NULL,
16     state VARCHAR(50) NOT NULL,
17     city VARCHAR(50) NOT NULL,
18     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
19     updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
20 );
21
22 CREATE TABLE passengers (
23     passenger_id INT PRIMARY KEY,
24     first_name VARCHAR(50) NOT NULL,
25     last_name VARCHAR(50) NOT NULL,
26     date_of_birth DATE NOT NULL,
27     gender VARCHAR(50) NOT NULL,
28     country_of_citizenship VARCHAR(50) NOT NULL,
29     country_of_residence VARCHAR(50) NOT NULL,
30     passport_number VARCHAR(20) NOT NULL UNIQUE,
31     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
32     updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
33 );
34
35 CREATE TABLE flights (
36     flight_id INT PRIMARY KEY,
37     sch_departure_time TIMESTAMP NOT NULL,
38     sch_arrival_time TIMESTAMP NOT NULL,
39     departing_airport_id INT NOT NULL,
40     arriving_airport_id INT NOT NULL,
41     departing_gate VARCHAR(50) NOT NULL,
42     arriving_gate VARCHAR(50) NOT NULL,
43     airline_id INT NOT NULL,
44     act_departure_time TIMESTAMP NOT NULL,
45     act_arrival_time TIMESTAMP NOT NULL,
46     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
47     updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
48 );
```

Data Output Messages Notifications

Commands completed successfully.
Total query runtime: 00:00:00.087
Rows affected: 0

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SQL screenshot 2

Query Tool - airport_db/postgres@localhost:5432 - pgAdmin 4

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SQLQuery2.sql - Booking tables

```
1 CREATE TABLE booking (
2     booking_id INT PRIMARY KEY,
3     flight_id INT NOT NULL,
4     passenger_id INT NOT NULL,
5     booking_platform VARCHAR(50) NOT NULL,
6     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
7     updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
8     status VARCHAR(50) NOT NULL,
9     price DECIMAL(7,2) NOT NULL
10 );
11
12 CREATE TABLE baggage_check (
13     baggage_check_id INT PRIMARY KEY,
14     check_result VARCHAR(50) NOT NULL,
15     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
16     updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
17     booking_id INT NOT NULL,
18     passenger_id INT NOT NULL
19 );
20
21 CREATE TABLE baggage (
22     baggage_id INT PRIMARY KEY,
23     weight_in_kg DECIMAL(4,2) NOT NULL,
24     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
25     updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
26     booking_id INT NOT NULL
27 );
28
29 CREATE TABLE boarding_pass (
30     boarding_pass_id INT PRIMARY KEY,
31     booking_id INT NOT NULL,
32     seat VARCHAR(50) NOT NULL,
33     boarding_time TIMESTAMP NOT NULL,
34     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
35     updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
36 );
37
38 CREATE TABLE booking_flight (
39     booking_flight_id INT PRIMARY KEY,
40     booking_id INT NOT NULL,
41     flight_id INT NOT NULL,
42     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
43     updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
44     CONSTRAINT uq_booking_flight UNIQUE (booking_id, flight_id)
45 );
46
47 CREATE TABLE security_check (
48     security_check_id INT PRIMARY KEY,
49     check_result VARCHAR(20) NOT NULL,
50     created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
51     updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
52     passenger_id INT NOT NULL
53 );
```

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SQL screenshot 3

Query Tool - airport_db/postgres@localhost:5432 - pgAdmin 4

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SQLQuery3.sql - Constraints

```
1  -- Required renames and structural changes
2  ALTER TABLE airline_info RENAME TO airline;
3  ALTER TABLE booking RENAME COLUMN price TO ticket_price;
4  ALTER TABLE flights ALTER COLUMN departing_gate TYPE TEXT;
5  ALTER TABLE airline DROP COLUMN info;
6
7  -- Relationships required by the assignment
8  ALTER TABLE security_check ADD CONSTRAINT fk_security_passenger
9      FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);
10 ALTER TABLE booking ADD CONSTRAINT fk_booking_passenger
11     FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);
12 ALTER TABLE booking ADD CONSTRAINT fk_booking_flight_direct
13     FOREIGN KEY (flight_id) REFERENCES flights(flight_id);
14 ALTER TABLE baggage_check ADD CONSTRAINT fk_baggage_check_booking
15     FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
16 ALTER TABLE baggage_check ADD CONSTRAINT fk_baggage_check_passenger
17     FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);
18 ALTER TABLE baggage ADD CONSTRAINT fk_baggage_booking
19     FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
20 ALTER TABLE boarding_pass ADD CONSTRAINT fk_boarding_pass_booking
21     FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
22 ALTER TABLE booking_flight ADD CONSTRAINT fk_booking_flight_booking
23     FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
24 ALTER TABLE booking_flight ADD CONSTRAINT fk_booking_flight_flight
25     FOREIGN KEY (flight_id) REFERENCES flights(flight_id);
26 ALTER TABLE flights ADD CONSTRAINT fk_flights_departing_airport
27     FOREIGN KEY (departing_airport_id) REFERENCES airport(airport_id);
28 ALTER TABLE flights ADD CONSTRAINT fk_flights_arriving_airport
29     FOREIGN KEY (arriving_airport_id) REFERENCES airport(airport_id);
30 ALTER TABLE flights ADD CONSTRAINT fk_flights_airline
31     FOREIGN KEY (airline_id) REFERENCES airline(airline_id);
32
33 -- Basic domain checks
34 ALTER TABLE baggage ADD CONSTRAINT ck_baggage_weight
35     CHECK (weight_in_kg > 0);
36 ALTER TABLE booking ADD CONSTRAINT ck_booking_ticket_price
37     CHECK (ticket_price >= 0);
38 ALTER TABLE flights ADD CONSTRAINT ck_flights_airports_different
39     CHECK (departing_airport_id <> arriving_airport_id);
```

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Rows affected: 0

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SQL screenshot 4

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SQLQuery4.sql - Master data

```
1  -- Airlines and airports
2  INSERT INTO airline
3  (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
4  VALUES
5  (1, 'KC', 'Air Astana', 'Kazakhstan', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
6  (2, 'DV', 'SCAT Airlines', 'Kazakhstan', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
7  (3, 'TK', 'Turkish Airlines', 'Turkey', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
8
9  INSERT INTO airport
10 (airport_id, airport_name, country, state, city, created_at, updated_at)
11 VALUES
12 (1, 'Almaty International Airport', 'Kazakhstan', 'Almaty', 'Almaty', CURRENT_TIMESTAMP,
13  CURRENT_TIMESTAMP),
14 (2, 'Nursultan Nazarbayev International Airport', 'Kazakhstan', 'Astana', 'Astana',
15  CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
16 (3, 'Shymkent International Airport', 'Kazakhstan', 'Shymkent', 'Shymkent', CURRENT_TIMESTAMP,
17  CURRENT_TIMESTAMP),
18 (4, 'Istanbul Airport', 'Turkey', 'Istanbul', 'Istanbul', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
19
20 INSERT INTO passengers
21 (passenger_id, first_name, last_name, date_of_birth, gender,
22  country_of_citizenship, country_of_residence, passport_number,
23  created_at, updated_at)
24 VALUES
25 (1, 'Amina', 'Sarsenova', '2001-04-12', 'Female', 'Kazakhstan', 'Kazakhstan', 'N100001',
26  CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
27 (2, 'Daniyar', 'Tulegen', '1999-11-03', 'Male', 'Kazakhstan', 'Kazakhstan', 'N100002',
28  CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
29 (3, 'Sofia', 'Lee', '2000-07-25', 'Female', 'Kazakhstan', 'Kazakhstan', 'N100003', CURRENT_TIMESTAMP,
30  CURRENT_TIMESTAMP),
31 (4, 'Noah', 'Kim', '1998-02-17', 'Male', 'Korea', 'Kazakhstan', 'N100004', CURRENT_TIMESTAMP,
32  CURRENT_TIMESTAMP),
33 (5, 'Aisha', 'Omar', '2002-09-09', 'Female', 'Kazakhstan', 'Kazakhstan', 'N100005',
34  CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
```

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SQL screenshot 5

Query Tool - airport_db/postgres@localhost:5432 - pgAdmin 4

File Edit View Query Tools Help

Open Save Execute Stop Database: airport_db

SQLQuery5.sql - Bookings

```
1  -- Flights and five normalized bookings
2  INSERT INTO flights
3  (flight_id, sch_departure_time, sch_arrival_time, departing_airport_id,
4   arriving_airport_id, departing_gate, arriving_gate, airline_id,
5   act_departure_time, act_arrival_time, created_at, updated_at)
6  VALUES
7  (101, '2026-10-01 08:00', '2026-10-01 09:40', 1, 2, 'A1', 'B2', 1, '2026-10-01 08:05', '2026-10-01
8   09:43', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
9  (102, '2026-10-02 14:00', '2026-10-02 15:20', 2, 3, 'C3', 'A2', 2, '2026-10-02 14:03', '2026-10-02
10  15:18', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
11  (103, '2026-10-03 06:30', '2026-10-03 11:40', 1, 4, 'D5', 'F1', 3, '2026-10-03 06:35', '2026-10-03
12  11:44', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
13
14 INSERT INTO booking
15 (booking_id, flight_id, passenger_id, booking_platform, created_at,
16  updated_at, status, ticket_price)
17 VALUES
18 (1001, 101, 1, 'Website', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 'CONFIRMED', 45000.00),
19 (1002, 101, 2, 'Mobile app', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 'CONFIRMED', 45000.00),
20 (1003, 102, 3, 'Travel agency', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 'CONFIRMED', 32000.00),
21 (1004, 103, 4, 'Website', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 'CONFIRMED', 185000.00),
22 (1005, 102, 5, 'Mobile app', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 'CHECKED_IN', 33500.00);
23
24 INSERT INTO booking_flight
25 (booking_flight_id, booking_id, flight_id, created_at, updated_at)
26 VALUES
27 (1, 1001, 101, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
28 (2, 1002, 101, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
29 (3, 1003, 102, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
30 (4, 1004, 103, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
31 (5, 1005, 102, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
32
33 INSERT INTO boarding_pass
34 (boarding_pass_id, booking_id, seat, boarding_time, created_at, updated_at)
35 VALUES
36 (1, 1001, '12A', '2026-10-01 07:20', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
37 (2, 1002, '12B', '2026-10-01 07:20', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
38 (3, 1003, '08C', '2026-10-02 13:20', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
39 (4, 1004, '21F', '2026-10-03 05:40', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
40 (5, 1005, '09A', '2026-10-02 13:20', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
```

Data Output Messages Notifications

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Rows affected: 0

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SQL screenshot 6

Query Tool - airport_db/postgres@localhost:5432 - pgAdmin 4

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Open Save Execute Stop Database: airport_db

SQLQuery6.sql - Checks

```
1  INSERT INTO baggage
2  (baggage_id, weight_in_kg, created_at, updated_at, booking_id)
3  VALUES
4  (1, 18.50, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1001),
5  (2, 20.10, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1002),
6  (3, 12.75, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1003),
7  (4, 23.40, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1004),
8  (5, 15.00, CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1005);
9
10 INSERT INTO security_check
11 (security_check_id, check_result, created_at, updated_at, passenger_id)
12 VALUES
13 (1, 'PASSED', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1),
14 (2, 'PASSED', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 2),
15 (3, 'PASSED', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 3),
16 (4, 'MANUAL_REVIEW', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 4),
17 (5, 'PASSED', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 5);
18
19 INSERT INTO baggage_check
20 (baggage_check_id, check_result, created_at, updated_at, booking_id, passenger_id)
21 VALUES
22 (1, 'CLEARED', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1001, 1),
23 (2, 'CLEARED', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1002, 2),
24 (3, 'CLEARED', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1003, 3),
25 (4, 'INSPECTION', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1004, 4),
26 (5, 'CLEARED', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP, 1005, 5);
27
28 -- Verification query
29 SELECT b.booking_id,
30        p.first_name || ' ' || p.last_name AS passenger,
31        bp.seat, a.airline_name,
32        da.city AS departure_city, aa.city AS arrival_city,
33        b.ticket_price
34 FROM booking b
35 JOIN passengers p ON p.passenger_id = b.passenger_id
36 JOIN boarding_pass bp ON bp.booking_id = b.booking_id
37 JOIN flights f ON f.flight_id = b.flight_id
38 JOIN airline a ON a.airline_id = f.airline_id
39 JOIN airport da ON da.airport_id = f.departing_airport_id
40 JOIN airport aa ON aa.airport_id = f.arriving_airport_id
41 ORDER BY b.booking_id;
42
43 -- Expected rows: 1001 Amina Sarsenova 12A Air Astana Almaty Astana 45000.00
44 --                  1002 Daniyar Tulegen 12B Air Astana Almaty Astana 45000.00
45 --                  1003 Sofia Lee      08C SCAT Airlines Astana Shymkent 32000.00
46 --                  1004 Noah Kim       21F Turkish Airlines Almaty Istanbul 185000.00
47 --                  1005 Aisha Omar     09A SCAT Airlines Astana Shymkent 33500.00
```

Data Output Messages Notifications

Query returned successfully.
Total query runtime: 00:00:00.087
Rows affected: 0

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